Machine Learning Classification of CHIME Fast Radio Bursts: II. Unsupervised Methods

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Abstract

Fast radio bursts (FRBs) are one of the most mysterious astronomical transients. Observationally, they can be classified into repeaters and apparently non-repeaters. This report summarizes the unsupervised machine learning methods applied to the first CHIME/FRB catalog to classify FRBs into clusters, revealing their differences and recommending future observational targets. The aim is to identify credible repeater candidates and evaluate algorithm performance in clustering high-dimensional data.

1 Introduction

1.1 Background

Fast Radio Bursts (FRBs) are enigmatic astrophysical phenomena characterized by intense bursts of radio waves lasting only milliseconds. First discovered in 2007 by the Parkes Telescope, FRBs are believed to originate from cosmological distances, making them valuable probes for studying the intergalactic medium, magnetic fields, and the underlying physics of their sources. Despite significant progress, the physical mechanisms responsible for FRBs remain largely unknown. The Canadian Hydrogen Intensity Mapping Experiment (CHIME) telescope has been instrumental in detecting a large number of FRBs and cataloging their properties.

1.2 Objective

The main goal of this work is to utilize unsupervised machine learning techniques to classify FRBs detected by CHIME into meaningful groups without relying on prior labels. By analyzing patterns in their observed and derived properties, this study aims to uncover potential subcategories, identify repeating FRBs, and propose candidates for follow-up observations.

1.3 Motivation

While traditional classification methods rely heavily on labeled data, many FRBs lack clear distinctions due to observational constraints. Unsupervised machine learning offers a way to analyze large datasets and reveal hidden structures, enabling researchers to go beyond human-imposed categories. With CHIME providing the largest FRB catalog to

date, leveraging these techniques can advance our understanding of FRBs, differentiate between repeaters and non-repeaters, and guide future observation strategies.

2 Data

The data used in this analysis were derived from the first CHIME/FRB catalog. This catalog includes 536 FRBs, out of which 474 are from non-repeating sources, and 62 are from 18 repeating FRB sources. Features were chosen based on their ability to differentiate between repeaters and non-repeaters. To ensure the best performance of our machine learning algorithms, I choose the features that show different distributions between repeaters and non-repeaters. Among the observed properties that can be directly included from the CHIME catalog, we employ peak frequency, flux, fluence, and boxcar width. Other physical properties including redshift, rest-frame frequency width, rest-frame width, energy, luminosity, and brightness temperature, are derived based on the directly observed properties. A brief description of these features is provided below. Many of the features, both from observed properties and physical properties, are logarithmically transformed, including flux, fluence, boxcar width, rest-frame shift, energy, and luminosity. The dataset used in this project is the file ($cul_cat1.csv$), which contains all the necessary logarithmic values.

2.1 Features

2.1.1 Observed Properties

- Peak Frequency (ν_c)
- Flux (S_ν)
- Fluence (F_ν)
- Boxcar Width (Δt_{BC})

2.1.2 Physical Properties

- Redshift (z)
- Rest-frame Frequency Width ($\Delta\nu$)
- Rest-frame Width (Δt_r)
- Energy (E)
- Luminosity (L)
- Brightness Temperature (T_B)

3 Methods

In this project, two unsupervised machine learning methods, clustering and dimensionality reduction, were applied.

3.1 Dimensionality Reduction Methods

Dimensionality reduction is the process of reducing the number of features or variables in a dataset while retaining as much of the meaningful information as possible. This is particularly useful for visualizing high-dimensional data or for preprocessing before applying other machine learning algorithms.

3.1.1 Principal Component Analysis (PCA)

- A linear dimensionality reduction technique.
- Identifies the directions (principal components) in which the variance of the data is maximized.
- Projects high-dimensional data onto a lower-dimensional space.
- Useful for reducing noise and visualizing data.
- Results are interpretable, as it ranks components by their explained variance.

3.1.2 t-distributed Stochastic Neighbor Embedding (t-SNE)

- A nonlinear dimensionality reduction technique.
- Preserves the local structure of data by mapping similar points in the high-dimensional space to be close in the lower-dimensional space.
- Well-suited for visualizing complex relationships in data, such as clusters.
- Computationally intensive and can struggle with large datasets.

3.1.3 Uniform Manifold Approximation and Projection (UMAP)

- Another nonlinear technique similar to t-SNE but faster and more scalable.
- Preserves both local and some global structure of the data.
- Based on mathematical foundations in Riemannian geometry and algebraic topology.
- Often preferred over t-SNE for large datasets or when retaining global structure is important.

3.2 Clustering Methods

Clustering involves grouping data points into clusters based on their similarity. These methods aim to reveal natural groupings in the data without prior knowledge of labels.

3.2.1 k-means Clustering

- Partitions data into k clusters, where k is predefined.
- Uses a distance metric (e.g., Euclidean distance) to assign points to the nearest cluster center (centroid).
- Iteratively updates cluster centroids to minimize the within-cluster variance.
- Works best for spherical clusters and may struggle with irregularly shaped clusters.

3.2.2 Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN)

- A density-based clustering algorithm.
- Improves upon DBSCAN by handling clusters with varying densities and incorporating a hierarchical approach.
- Automatically determines the number of clusters.
- Robust to noise and irregularly shaped clusters, making it suitable for complex datasets.
- Can also identify "noise points" (outliers) that do not belong to any cluster.

4 Results

4.1 PCA with k-means

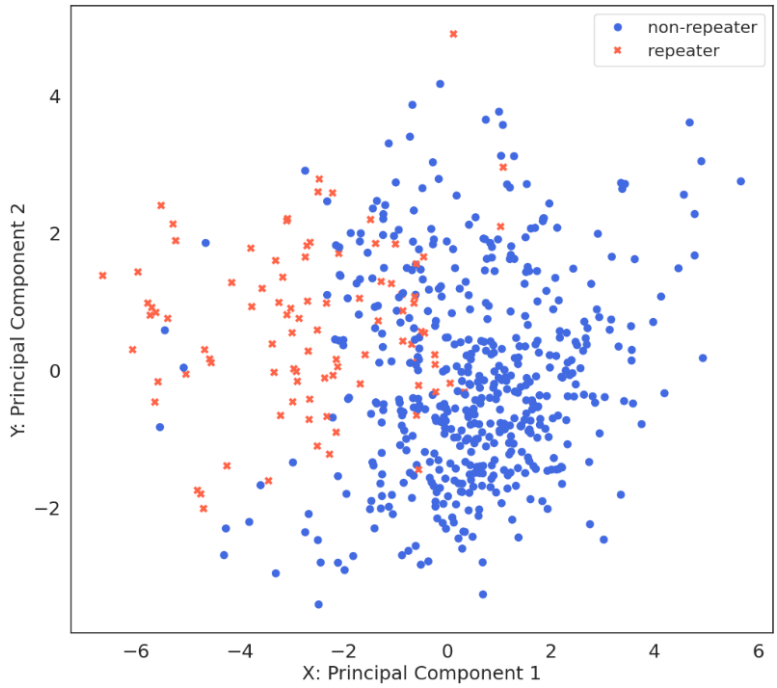

Figure 1: Dimensionality reduction results from PCA. The samples of repeating FRBs and non-repeating FRBs are clearly distributed in different regions

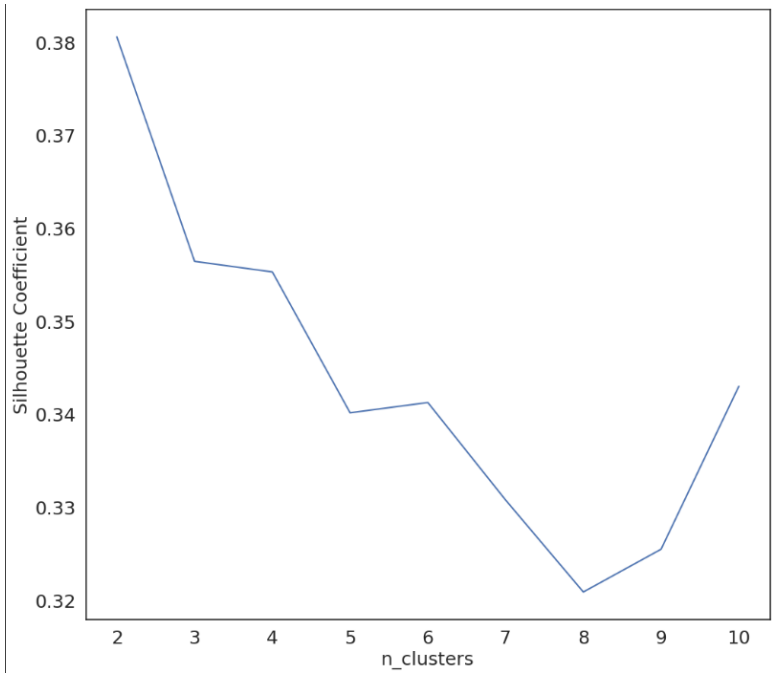

Figure 2: Silhouette coefficients of k-means after PCA with respect to different number of clusters. The maximum value of silhouette coefficient implies 2 clusters in PCA space and the corresponding hyperparameter for k-means.

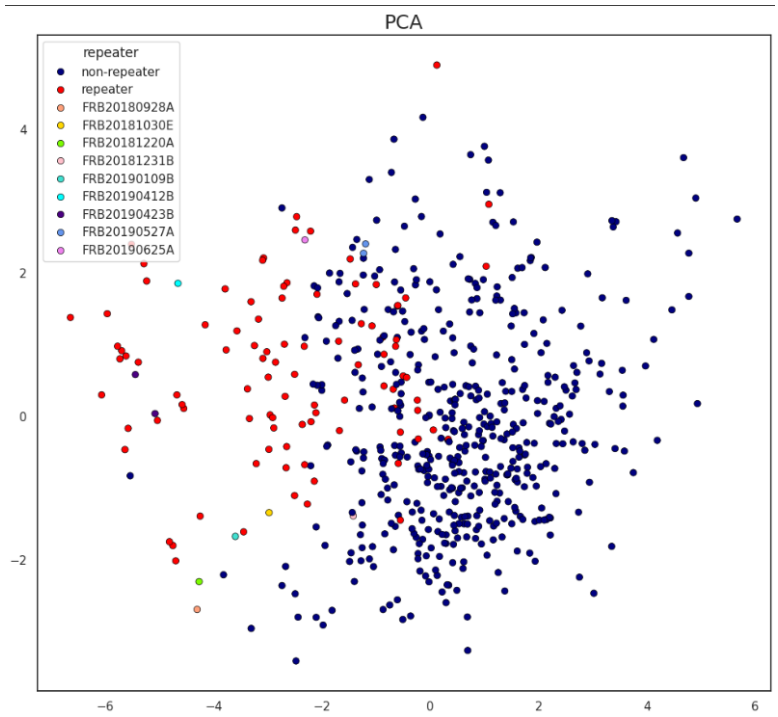

Figure 3: The k-means clustering result in PCA space. The right cluster, containing a higher ratio of repeaters, is identified as the repeater cluster, while the left one is identified as the non-repeater cluster. The two clusters are consistent with the distribution of observed repeaters and non-repeaters in Figure 1.

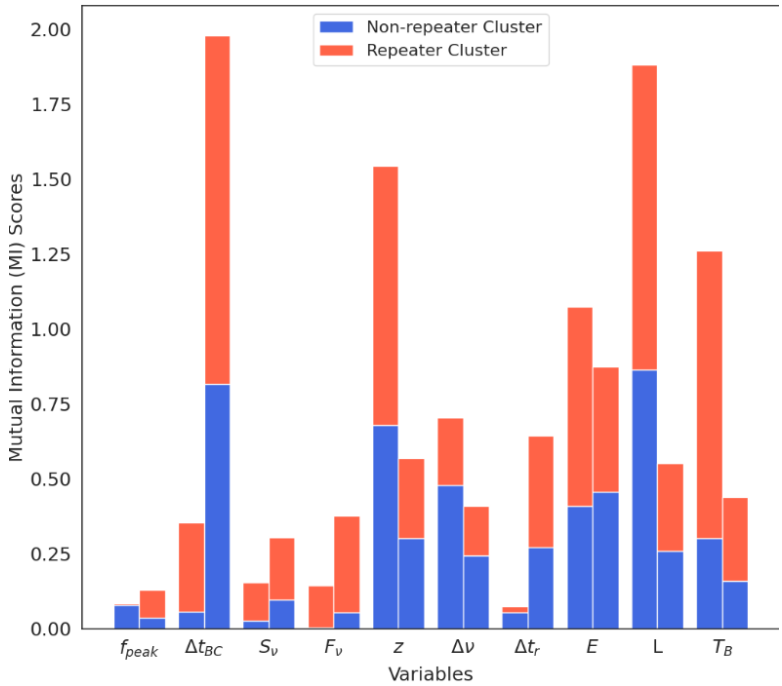

Figure 4: . Feature correlation of PCA+k-means. The left bar correlates with the x-axis while the right bar is related to the y-axis. Boxcar width, redshift, energy, luminosity and brightness temperature are the most important features in PCA.

4.2 t-SNE with HDBSCAN

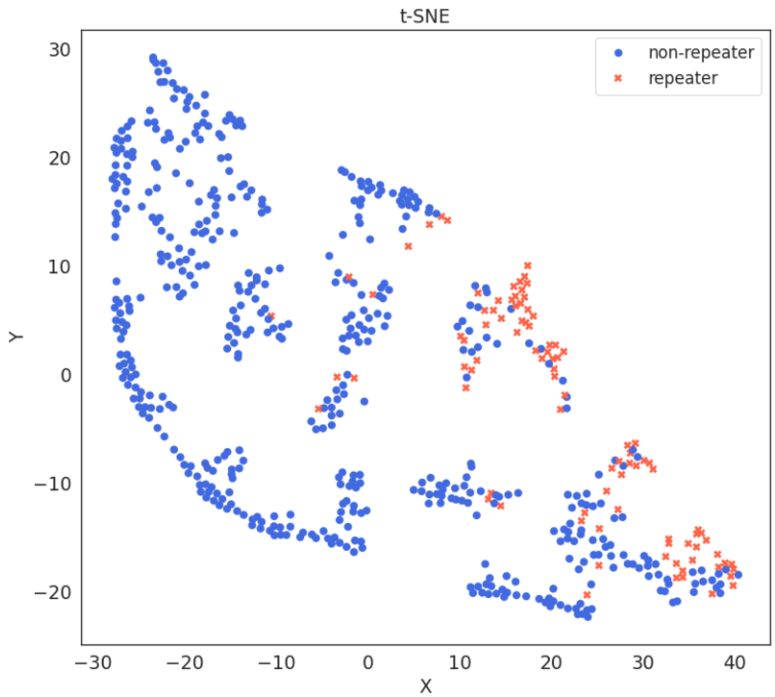

Figure 5: Dimensionality reduction results from t-SNE. Most repeating FRBs are on the right-hand side and generally reside in the same clusters.

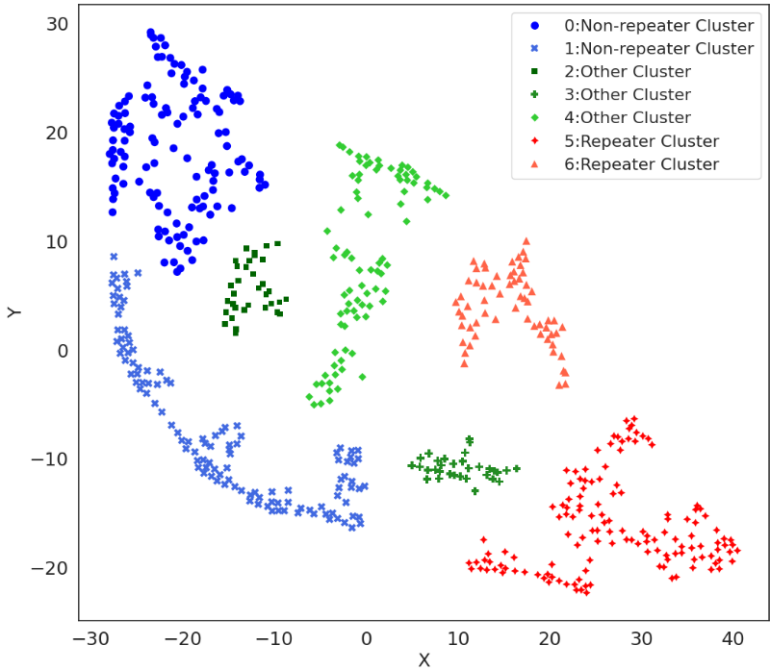

Figure 6: Clustering results of HDBSCAN in the t-SNE plane. Non-repeater clusters are marked in blue, while repeater clusters are marked in red. Other clusters are marked in green.

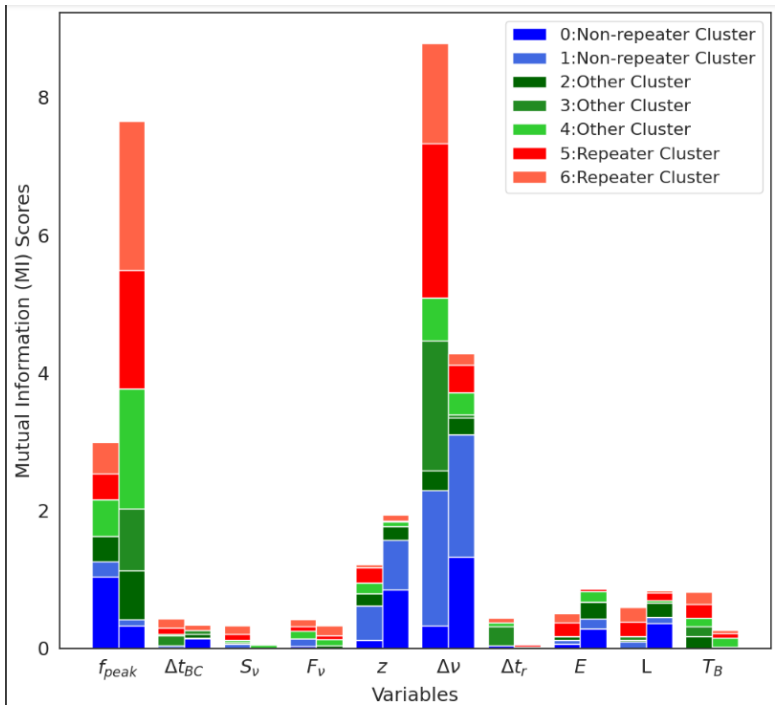

Figure 7: Feature correlation of t-SNE+HDBSCAN. For nonrepeater clusters in blue, peak frequency, redshift and rest-frame frequency width are the most dominant features, while redshift is not as significant in the repeater clusters colored red. On the whole, peak frequency, redshift and rest-frame frequency width are the most important features.

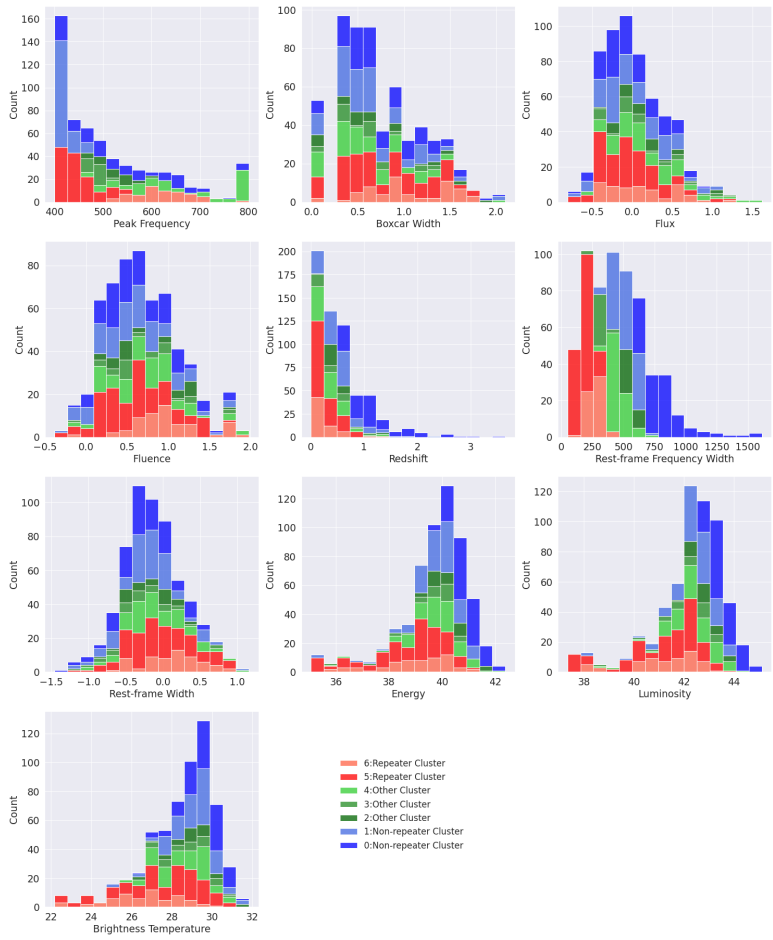

Figure 8: . Input features of FRBs in the labels given by t-SNE and HDBSCAN. The different distributions of the clusters imply the corresponding categories

4.3 UMAP with HDBSCAN

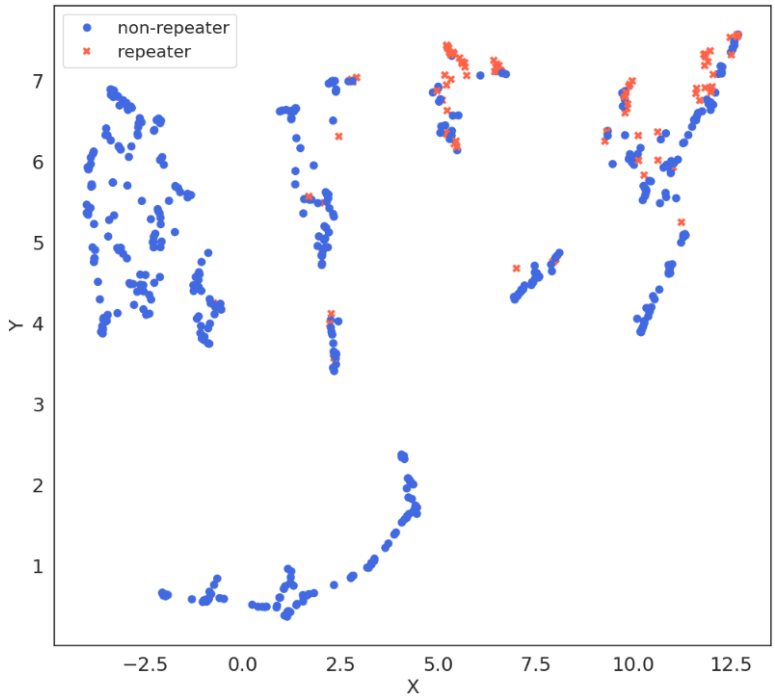

Figure 9: Dimensionality reduction results from UMAP. Similar to t-SNE, most repeating FRBs are on the right-hand side and generally reside in the same clusters.

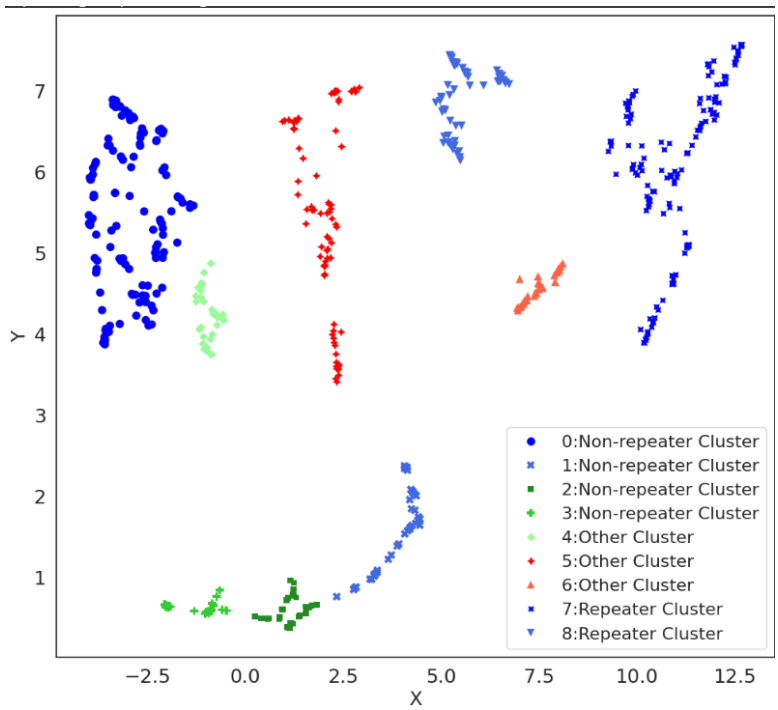

Figure 10: Clustering results of HDBSCAN in the UMAP plane. Non-repeater clusters are marked in blue, while repeater clusters are marked in red. Other clusters are marked in green.

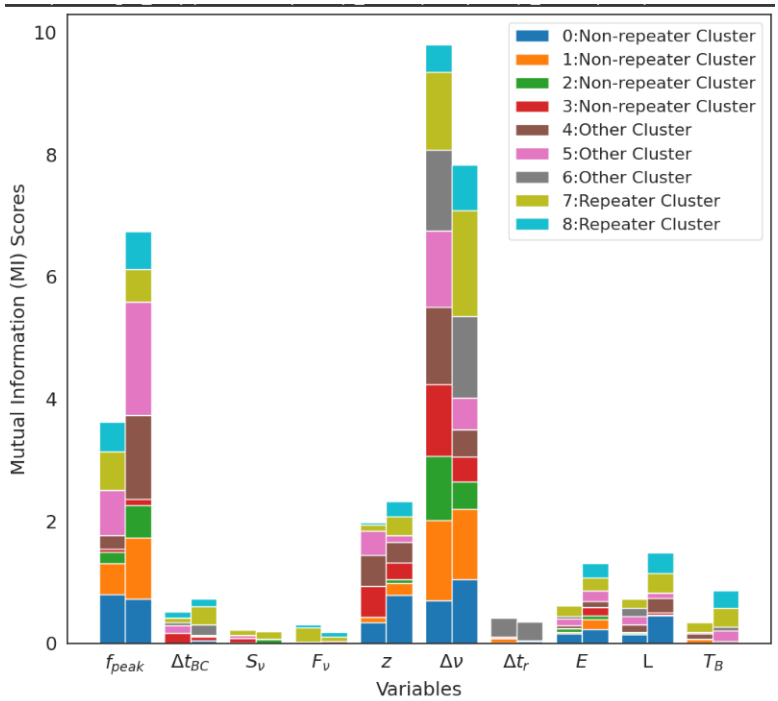

Figure 11: . Features correlation of UMAP+HDBSCAN. For nonrepeater clusters in blue, peak frequency, redshift and rest-frame frequency width are the most dominant features, while redshift is not as significant in the repeater clusters colored red. On the whole, peak frequency, redshift and rest-frame frequency width are the most important features.

5 Discussions

The results suggest that repeaters and non-repeaters are physically distinct categories. Additional subclusters indicate possible variations within these primary classes. Dimensionality reduction and clustering provide valuable tools for uncovering these hidden structures.

6 Conclusion

- Unsupervised machine learning effectively identifies repeater candidates from apparently non-repeating FRBs. Future observations of these candidates will help refine classification methods and improve our understanding of FRBs.
- In addition to spotting the two most significant categories of repeaters and non-repeaters, unsupervised learning methods also imply that there may exist 5 to 7 subspecies based on their traits.

Data Availability

The data used in this project work is available at `https://www.chime-frb.ca/catalog`.

Project Files

The codes and other supplementary material are available at `https://github.com/Peeresh/Project-files`.

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